

R-Squared as a Sample Statistic

Teaching note

Brown, Lo, and Lys (1999), *Journal of Accounting and Economics*
Gu (2007), *Journal of Business Finance & Accounting*

ACCTG 597A · Fall 2026 · Sam Bonsall · Penn State

R^2 differences were read as economic change

“Frequently, the studies operationalize value relevance as the R^2 from regressions of stock prices on per share values of accounting earnings and book values of equity. Inter-temporal or cross-sample differences in R^2 are used as indicators that value relevance of accounting disclosures has changed over time or that value relevance differs across disclosure regimes.”

Brown, Lo, and Lys (1999, p. 84)

Brown, Lo, and Lys (BLL): R^2 depends on the coefficient of variation of unobserved scale; controlling it restores comparability.

Gu (2007): R^2 contains the sample variance of x regardless of scale.

Both claims follow from one identity.

BLL reverse CMW's trend with a scale control

BLL replicate Collins, Maydew, and Weiss (1997, CMW) by year (Eq. (9)), then regress the 39 annual R^2 s on time and coefficient-of-variation (CV) proxies for scale (Eq. (14); pp. 93, 98):

$$P_{i,t} = a_{0,t} + a_{1,t} EPS_{i,t} + a_{2,t} BVPS_{i,t} + e_{1,i,t} \quad (9)$$

$$R_t^2 = g_0 + g_1 Time + g_2 \hat{\omega}_{P,t} + g_3 \hat{\omega}_{BVPS,t} + e_{5,t} \quad (14)$$

Model (BLL Table 4, Panel A)	<i>Time</i>	$\hat{\omega}_{BVPS,t}$
1: time only	+0.002 (1.567)	
3: CV proxy only		0.479 (5.761)
5: time and CV proxy	-0.003 (-2.051)	0.612 (5.961)
6: time and both CV proxies	-0.003 (-1.994)	0.630 (3.902)

Recreated from BLL Table 4, Panel A (WLS, $T = 39$, p. 102); t -statistics in parentheses.
Model 6: $\hat{\omega}_{P,t} -0.025 (-0.152)$.

With the CV proxies in the regression, the Time coefficient changes sign.

The abstract's condition: comparisons are invalid "unless one controls for differences in the scale factor's coefficient of variation" (p. 83).

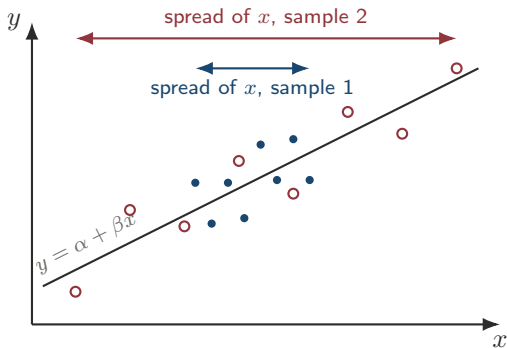
Widening x alone raises R^2

Every observation obeys the same relation; only the spread of x differs.

Same line $y = \alpha + \beta x$, same eight residuals, in both samples (adapted from Gu, Figure 1a, p. 1078).

Sample 1: narrow range of x ; most variation in y is residual, so R^2 is low.

Sample 2: wide range of x ; the same residuals are small relative to the line's variation, so R^2 is high.



R^2 depends on σ_x^2 , which no theory restricts

Gu's relation (Eq. (1), p. 1075) and the definition of R^2 (p. 1076), with e_i the OLS residuals:

$$y_i = \alpha + \beta x_i + \varepsilon_i, \quad \text{Var}(\varepsilon_i) = \sigma_\varepsilon^2 \quad (1) \quad R^2 = 1 - \frac{\sum_i e_i^2}{\sum_i (y_i - \bar{y})^2}$$

Substitute $\sum_i (y_i - \bar{y})^2 = \hat{\beta}^2 \sum_i (x_i - \bar{x})^2 + \sum_i e_i^2$ (OLS orthogonality), divide by $n - 1$, and take limits:

$$R^2 = \frac{\hat{\beta}^2 \hat{\sigma}_x^2}{\hat{\beta}^2 \hat{\sigma}_x^2 + \hat{\sigma}_\varepsilon^2}, \quad \text{plim } R^2 = \frac{\beta^2 \sigma_x^2}{\beta^2 \sigma_x^2 + \sigma_\varepsilon^2}, \quad \frac{\partial \text{plim } R^2}{\partial \sigma_x^2} = \frac{\beta^2 \sigma_\varepsilon^2}{(\beta^2 \sigma_x^2 + \sigma_\varepsilon^2)^2} > 0$$

$\hat{\beta} \rightarrow \beta$ and $\hat{\sigma}_\varepsilon^2 \rightarrow \sigma_\varepsilon^2$: parameters of the relation

$\hat{\sigma}_x^2$: a property of the sample

Two samples share a population R^2 only if “both the underlying population is held constant and the samples are drawn randomly from the same population” (p. 1077).

Simulated: $\hat{\beta}$ and $\hat{\sigma}_\varepsilon$ agree, R^2 does not

```
set seed 1999
set obs 100000
gen x = 2 + rnormal() // sample 1
gen e = rnormal()
gen y = x + e // beta = 1
reg y x
// sample 2: sd(x) = sqrt(3)
// R2 = b^2 s2x/(b^2 s2x+s2e)
```

	Sample 1	Sample 2
$\hat{\beta}$ (s.e.)	1.00 (0.003)	1.00 (0.002)
$\hat{\sigma}_\varepsilon$ (residual s.d.)	1.00	1.00
$\hat{\sigma}_x$	1.00	1.74
R^2	0.50	0.75
plim R^2	0.50	0.75

Sample 1: $x \sim N(2, 1)$; sample 2: $x \sim N(2, 3)$;
 $y = x + \varepsilon$, $\varepsilon \sim N(0, 1)$; seed 1999, $N = 100,000$ per
sample;
scripts/bll_r2_incomparability_twosample.do;
the log verifies both identities to eight decimals.

The maroon cells differ only because $\hat{\sigma}_x$ differs.

BLL Table 5 still compares R^2 across samples

Deflating by $P_{i,t-1}$ (Eq. (15), p. 105) removes scale if P_{t-1} proxies s , yet the 39 annual R^2 s of BLL Table 5 come from cross-sections whose spread of EPS/P_{t-1} and $BVPS/P_{t-1}$ differs.

By the identity, the decline is consistent with a larger σ_ε^2 (BLL's reading), a smaller β , or a changed dispersion or covariance of the regressors; R^2 cannot say which.

With the outlier filter iterated (Gu fn. 16, p. 1086): "Brown et al.'s finding of a decreasing trend in the R^2 s for the returns model (3) is not robust".

Trend in returns-model R^2 , OLS	t -statistic
BLL Table 5, Panel B, 1958–1996 (p. 106)	−2.084
Gu fn. 16: single-pass outlier removal	−2.030
Gu fn. 16: iterated removal, 1958–1996	−0.890
Gu fn. 16: iterated removal, 1953–1998	−1.646

At $\alpha = 0$, scale raises R^2 by widening x

$$z_i = \alpha + \beta w_i + \varepsilon_i \quad (1) \quad y_i = s_i z_i, \quad x_i = s_i w_i \quad (2) \quad y_i = b_0 + b_1 x_i + \eta_i \quad (4)$$

BLL, pp. 87–89. With $\alpha = 0$ (the case of Eq. (8')) and s independent of w and z , hence of ε (assumption 3), $y_i = \beta x_i + s_i \varepsilon_i$ and

$$\begin{aligned} \text{Var}(x) &= \mu_s^2 D, & \text{Var}(s\varepsilon) &= \mu_s^2 (1 + \omega_s^2) \sigma_\varepsilon^2, & D &= \omega_s^2 \sigma_w^2 + \omega_s^2 \mu_w^2 + \sigma_w^2 \quad (\text{Eq. (7)}) \\ \text{plim } R^2 &= \frac{\beta^2 \text{Var}(x)}{\beta^2 \text{Var}(x) + \text{Var}(s\varepsilon)} = \frac{\beta^2 V}{\beta^2 V + \sigma_\varepsilon^2}, & V &\equiv \frac{D}{1 + \omega_s^2} = \sigma_w^2 + \mu_w^2 \frac{\omega_s^2}{1 + \omega_s^2} \\ \frac{\partial \text{plim } R^2}{\partial \omega_s^2} &= \frac{\beta^2 \sigma_\varepsilon^2 \mu_w^2}{A^2}, & A &\equiv \omega_s^2 \sigma_z^2 + \omega_s^2 \mu_z^2 + \sigma_z^2 = (1 + \omega_s^2)(\beta^2 V + \sigma_\varepsilon^2) \quad (\text{Eq. (8')}, \text{ p. 90}) \end{aligned}$$

$\omega_s = 0$: $V = \sigma_w^2$, the scale-free R^2 .

$\omega_s \rightarrow \infty$: $V \rightarrow \sigma_w^2 + \mu_w^2$, the uncentered R^2 of z on w ; it nears one only when the intercept dominates, BLL's "essentially regressing s on itself" (p. 91).

Scale enters R^2 only through V : BLL's scale effect is Gu's σ_x^2 effect with one added term (derived here).

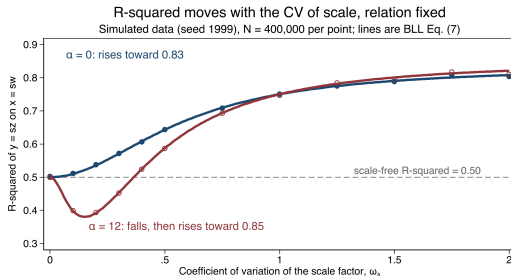
The CV effect on R^2 can change sign

$\alpha = 0$: the curve rises from 0.50 toward 0.83, the uncentered R^2 of z on w (Eq. (8')).

$\alpha = 12$: the derivative is negative at $\omega_s = 0$ and turns positive near 0.15; R^2 falls to about 0.38 and stays below the scale-free 0.50 until $\omega_s \approx 0.37$.

For $\alpha > 0$, R^2 need not rise with ω_s : neither p. 91 conclusion holds.

The erratum concedes: “we cannot rule out the possibility that the derivative may change sign as ω_s increases” (p. 141).



Markers: simulated R^2 of $y = sz$ on $x = sw$; lines: Eq. (7). $w \sim N(2, 1)$, $\varepsilon \sim N(0, 1)$, $\beta = 1$, s lognormal with unit mean; `bll_r2_incomparability_scale_sweep.do`.

The CV control leaves the σ_w^2 channel open

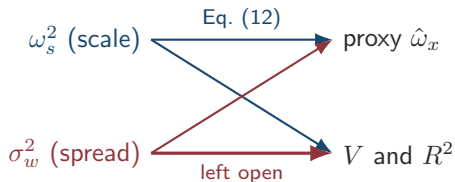
BLL's proxies are the CVs of observed price and BVPS; with s and w independent (Eq. (12), p. 96),

$$\omega_x^2 = \omega_s^2 (1 + \omega_w^2) + \omega_w^2$$

so the proxy moves with ω_w as well as ω_s (fn. 6), while V , hence R^2 , moves with σ_w^2 whatever ω_s is.

Even a perfect control for ω_s leaves σ_w^2 inside V , and the linear control in Eq. (14) presumes a monotone relation.

The erratum retains the dependence: “our most important conclusion remains unchanged: R^2 functionally depends on ω_s ” (p. 141).



Residual variance ignores σ_x^2 , not scale

Gu's measure (p. 1080) estimates "an economic parameter of the economic relation that the researcher believes in" (p. 1079), yet under BLL's scale factor the residual variance rises with ω_s :

$$\hat{\sigma}_\varepsilon^2 = \frac{\sum_i e_i^2}{n - k}, \quad E(\hat{\sigma}_\varepsilon^2) = \sigma_\varepsilon^2 \quad \text{Var}(s\varepsilon) = \mu_s^2 (1 + \omega_s^2) \sigma_\varepsilon^2$$

	Baseline	(a) σ_w : 1 to 1.73	(b) ω_s : 0 to 1	(c) σ_ε : 1 to 1.73
R^2	0.50	0.75	0.75	0.25
$\hat{\beta}$	1.00	1.00	1.00	1.00
$\hat{\sigma}_\varepsilon$ (residual s.d.)	1.00	1.00	1.42	1.73
$\hat{\sigma}_x$	1.00	1.74	2.46	1.00

Seed 1999, N = 100,000 per sample; $y = sz$, $x = sw$, s lognormal with unit mean; scripts/bll_r2_incomparability_three_sources.do.

R^2 cannot separate (a) from (b); the residual s.d. can, but confounds (b) with (c): Gu's measure needs its own scale control.

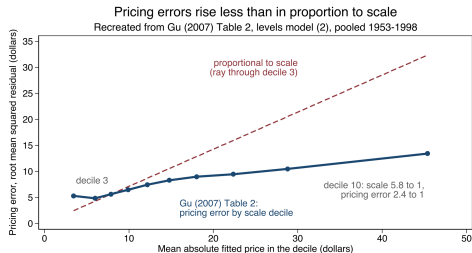
Pricing errors are not proportional to scale

Gu writes $\sigma_{\varepsilon}^2 = f(S) \sigma_v^2$ (p. 1081); a multiplicative s makes f proportional to scale squared.

Decile 10 has 5.8 times the scale of decile 3 but 2.4 times the pricing error (Gu Table 2).

“Stocks at \$40 per share do not have pricing errors twice as large as those for stocks at \$20 on average” (p. 1082).

Under decile benchmarks, levels-model abnormal pricing errors rise after 1973 (Gu Table 3, Panel A: 0.087 per year, $t = 4.238$, p. 1092).



Recreated from Gu Table 2 (p. 1089), levels model; `b11_r2_incomparability_gu_table2.do`. The ray puts decile 10's error at 32.4 against 13.4.

Every cross-sample R^2 test inherits σ_x^2

Gu (2007, fn. 2) lists twelve such studies (p. 1074); regressor dispersion presumably differs in each as well:

Setting	R^2 compared across	What also differs	Comparable statistic
Over time (CMW 1997; BLL 1999)	annual cross-sections	spread of <i>EPS</i> and <i>BVPS</i> ; CV of scale	residual variance, nonlinear scale control (Gu, Tables 3 to 5)
Across countries (Harris et al. 1994)	national samples	spread of the accounting variables; scale	residual variance within one model and one y
Across partitions (Basu 1997; Week 4 module)	subsamples split on the regressor	spread of the regressor, by construction	slope coefficients; residual variance by partition
Across dependent variables (levels, returns)	specifications	the variance of y itself	none; a Vuong test needs the same y (Gu fns. 5, 7)

With several regressors, “The joint effect on the R^2 is practically unpredictable” (p. 1077).

BLL's tables support dependence, not a remedy

Supported: R^2 depends on ω_s (Eq. (7)), and the BLL Table 4 trend is not robust to a nuisance control.

Not supported: the abstract's "unless"; the proxy carries ω_w , and σ_w^2 stays inside R^2 .

"Conversely, finding that R^2 is not significantly different also does not imply that value-relevance does not differ across the samples, because scale effects may offset differences in value relevance."

Brown, Lo, and Lys (1999, p. 92)

A skeptic would note, with Gu, that "If the model is regarded invalid, then no measure (R^2 or any other) will be meaningful" (p. 1079).

Summary

- R^2 combines two parameters of the economic relation, the slope and the error variance, with the sample dispersion of the regressor, which no theory restricts; two samples that obey the same relation can therefore differ in R^2 by any amount, so an R^2 difference is not, by itself, evidence of a different relation.
- The scale effect of Brown, Lo, and Lys appears to be a special case of this dependence: with no intercept in the scale-free relation, their probability limit reduces to Gu's form with an effective regressor variance $\sigma_w^2 + \mu_w^2 \omega_s^2 / (1 + \omega_s^2)$, so a control for the coefficient of variation of scale closes one channel and leaves the other open.
- The residual variance estimates a parameter of the relation and is invariant to the regressor's dispersion; comparisons of explanatory power across samples appear better grounded in it than in R^2 , given a scale control that does not presume proportionality. Under that measure, Gu documents declining value relevance since the early 1970s in every specification.

Brown, Lo, and Lys (1999) · Gu (2007) · Collins, Maydew, and Weiss (1997) ·
github.com/sbb151/acctg597a-research-design